



Telecom-wavelength conversion in a high optical depth cold atomic system

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Abstract: We experimentally investigate the frequency down-conversion through the four-wave mixing (FWM) process in a cold ⁸⁵Rb atomic ensemble, with a diamond-level configuration. An atomic cloud with a high optical depth (OD) of 190 is prepared to achieve a high efficiency frequency conversion. Here, we convert a signal pulse field (795 nm) attenuated to a single-photon level, into a telecom light at 1529.3 nm within near C-band range and the frequency-conversion efficiency can reach up to 32%. We find that the OD is an essential factor affecting conversion efficiency and the efficiency may exceed 32% with an improvement in the OD. Moreover, we note the signal-to-noise ratio of the detected telecom field is higher than 10 while the mean signal count is larger than 0.2. Our work may be combined with quantum memories based on cold ⁸⁵Rb ensemble at 795 nm and serve for long-distance quantum networks.

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1. Introduction

One of the most prominent and interesting phenomena of the interaction between optical fields and a nonlinear medium is the four-wave mixing (FWM) [1] that originates from the third-order nonlinear process. Recently, FWM processes have found various applications in atomic systems, such as the generation of correlated photon pairs [2,3] or squeezed light [4], the storage of coherent light [5], optical frequency conversion [6] and phase-insensitive optical amplification [7], etc. Among them, the long-distance quantum communication processing [8] is an extensively researched topic, which includes generating telecom-wavelength photons [9,10] or converting the frequency of photons into the telecom-wavelength window.

To date, frequency conversion based on FWM in the rubidium atomic system has been widely investigated in various energy-level schemes, such as ladder-type [11–13], double-lambda type [14,15], and diamond-type [16,17]. For example, double-lambda schemes with conversion efficiency up to 91.2% have been reported [18]. Unfortunately, the generated photons are still limited in the visible-wavelength window (from 780 nm to 795 nm wavelengths), which restricts its applications in long-distance communications due to the inevitable high losses in fibre-optical channels. One conventional way towards the conversion between photons in telecom and visible spectrum range is adopted to diamond schemes. Although an E-band [19] frequency conversion with an efficiency as high as 54% has been realized in the profound work recently [20], the converted photons (1367 nm) are still not in the best wavelength of a telecom-fiber. More specifically, a photon in 1367 nm has a much larger transmission loss of 0.37 dB/km, compared to that of the photon in 1529.3 nm (0.19 dB/km). For instance, when incorporated into a 100

km long-distance communication network, a converter with 1529.3 nm provides an advantage of 18 dB of less loss in the fiber. Until now, the frequency conversion process from 795 nm wavelength to 1529.3 nm (near C-band) wavelengths with a co-linear propagation structure of lights in a cold ^{85}Rb ensemble has not been reported, and we explored this conversion process in our experiment. As been studied in the pioneering work [21], the attainable conversion efficiency is highly related to the optical depth (OD) of the medium. In this work, we establish a dense cold ^{85}Rb atomic ensemble, the OD of which is measured to be 190 in an independent experiment, by utilizing a technique called dark-spontaneous emission optical trap [22] and then realize a telecom-wavelength conversion based on nonlinear optical frequency down-conversion process of FWM.

Here, we consider a diamond configuration ($5S_{1/2} - 5P_{1/2} - 4D_{3/2} - 5P_{3/2} - 5S_{1/2}$) in cold ^{85}Rb as it covers resonance frequency of telecom-wavelength (1529.3 nm) and a wavelength (795 nm) that commonly used for atom-based optical memory [23]. The frequency down-conversion process is investigated while the mean photon count of the input signal field is attenuated to the magnitude of a single photon level. As described by the theory in previous work [21], we experimentally verified that the frequency-conversion efficiency increases with the OD. Such frequency converters, operating at the single photon level and high OD, play an important role in atom-based quantum repeater and long-distance quantum communication.

2. Experimental setup

Figure 1(a) shows a four-levels diamond configuration of ^{85}Rb atomic system, which consists of two intermediate states $|2\rangle$ ($|5P_{1/2}, F' = 3\rangle$) and $|4\rangle$ ($|5P_{3/2}, F' = 4\rangle$), one ground state $|1\rangle$ ($|5S_{1/2}, F = 3\rangle$), and one excited state $|3\rangle$ ($|4D_{3/2}, F'' = 3\rangle$). We define the detuning Δ_s (Δ_{p2}) with the form of $\Delta_s = \omega_s - \omega_{12}$ ($\Delta_{p2} = \omega_{p2} - \omega_{23}$), where ω_s (ω_{p2}) is the frequency of signal (pump2) field and ω_{ij} is the frequency of the $|i\rangle \leftrightarrow |j\rangle$ transition. The signal (795 nm) and pump1 (1475 nm) fields couple the atomic transition $|1\rangle \rightarrow |2\rangle$ with a detuning of $\Delta_s = -2\pi \times 6$ MHz (The negative sign means red detuning,) and the atomic transition $|2\rangle \rightarrow |3\rangle$ under resonance, respectively. The transition between $|1\rangle \leftrightarrow |4\rangle$ is coupled by a pump2 (780 nm) field with a detuning of $\Delta_{p2} = -2\pi \times 36$ MHz. According to the principle of energy conservation, the telecom (FWM) light at 1529.3 nm can be generated through the transition from $|3\rangle$ to $|4\rangle$ in the FWM process, providing the phase-matching condition is satisfied.

A simplified schematic of our experiment is depicted in Fig. 1(b). The nonlinear media is a cigar-shaped ^{85}Rb atomic cloud (with an OD of 190) that is prepared in a dark-line two-dimensional magneto-optical trap (MOT). We focus strong pump1 (1.87 mW) and pump2 (1.4 mW) fields, together with a single-photon-level signal light into the ensemble with different waists of 152 μm , 117 μm , and 71 μm , respectively. The pump fields and signal light propagate collinearly in the ^{85}Rb atoms with vertical and horizontal polarization. We block the pump2 field with a dichroic mirror (DM2) and use interference filters (IF2) and a polarization beam splitter (PBS) to filter out the generated telecom light with horizontal polarization from the strong pump1 light. The telecom light is collected by a fiber collimator (FC2) and a fiber filter (FF) is used to isolate the background fluorescence and further improve the signal-to-noise ratio (SNR). Finally, we detect the generated telecom photon with an infrared single-photon detector (Qasky WT-SPD100, InGaAs Photon Detector, 10% detection efficiency) which is connected to the FF. Similarly, we use the IF1 to filter out the signal field and measure the count of input signal photons by an avalanche diode detector (PerkinElmer SPCM-AQR-16-FC, 65% detection efficiency). The time sequence is shown in Fig. 1(c). Our system is repeated with a period of 10 ms including an 8 ms trapping time and an experimental time of $\Delta T = 2$ ms (The MOT magnetic field and MOT beams are turned off.). In each cycle, we modulate the signal light and pump fields with the temporal modes of a Gaussian pulse and square-shaped pulses. The square-shaped pulses of the pump fields are used to keep the instantaneous intensities of the pump fields constant during

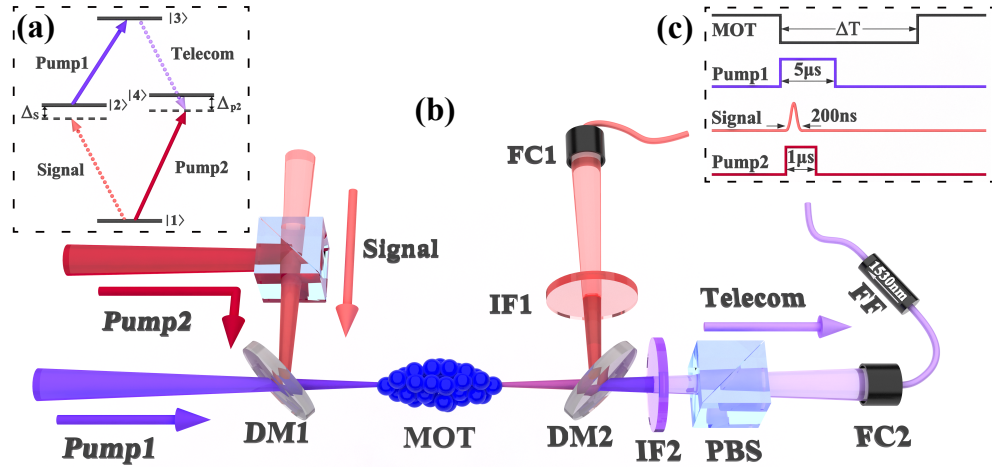


Fig. 1. (a) Energy-level diagram of diamond configuration. (b) Schematic diagram of the experimental setup. MOT: magneto-optical trap; PBS: polarizing beam splitter; DM1, DM2: dichroic mirror; IF1, IF2: Interference filter; FC1, FC2: fibre collimator; FF: fibre filter. (c) Time sequence for the frequency down-conversion process.

the FWM process, and the conversion efficiency will decrease when the pump intensities are weak. We have not observed a significant difference between the condition where the signal pulse is Gaussian-shaped and the condition of a square-shaped signal pulse. However, considering the need of transmitting the signal light in an optical fiber for further applications, a temporal mode of Gaussian pulse is chosen in our work. The pulse time width of pump1, pump2, and signal fields are 1 μ s, 5 μ s, and 200 ns respectively. The experimental times for the pumping fields (pump1 and pump2) in our work need to be less than 2 ms (the experimental window in each cycle) and should overlap with the signal pulse. The temporal width of the near-resonance pump2 light should be tailored to be as short as possible to avoid introducing an influence on the effective optical depth.

3. Experimental results and theoretical analysis

In our experiment, the input signal pulse with a temporal width of 200 ns and pulse repetition rate of $R_p = 100$ Hz, is attenuated to the single-photon level by a neutral density filter. By taking the transmittance (58.5%) of the uncoated glass vacuum cell and various optical components, the coupling efficiency (83.7%) of the FC1, and the detection efficiency (65%) of the detector into consideration, we can obtain the number of input signal photons through dividing the count of detected signal photons (N_s) by a factor of $T_s = 0.318$. Similarly, We can calculate the number of converted telecom photons by considering the count of detected telecom photons (N_t) with the transmission factor $T_t = 0.0299$, where T_t includes the transmittance (54.3%) of the MOT and various optical components, the detection efficiency (10%) of the detector, the transmittance (91.8%) of the FF, and the coupling efficiency (60%) of the FC2. Here, we define the frequency-conversion efficiency as:

$$\eta = N_t T_s / (N_s T_t). \quad (1)$$

In Fig. 2(a), the average counts of input signal photons per pulse is ensured to be $\bar{n} = 0.8$ ($\bar{n} = N_s / (300 R_p T_s)$) by a coincidence count of 300 s. We adjust the signal detuning Δ_s through an acousto-optic modulator (AOM), and the frequency-conversion efficiency η is measured while the Δ_s is scanned from $-2\pi \times 40$ MHz to $2\pi \times 40$ MHz. We find that η is higher in the case of a

near-resonance signal field, and the main reasons include the strong atomic absorption [20] and the phase-matching condition satisfied by the frequencies of the experimental beams in our work. The solid line is fitted by the Lorentzian function $y = y_0 + 2Aw/(\pi(4(x - x_c)^2 + w^2))$ with fitted values ($y_0 = -0.306$, $x_c = -1.507$, $w = 125.9$, $A = 119.8$). The fitting curve gives a bandwidth of 70 MHz in FWHM. We find that the frequency-conversion efficiency of our system reaches its maximum value when the signal detuning Δ_s is $-2\pi \times 6$ MHz.

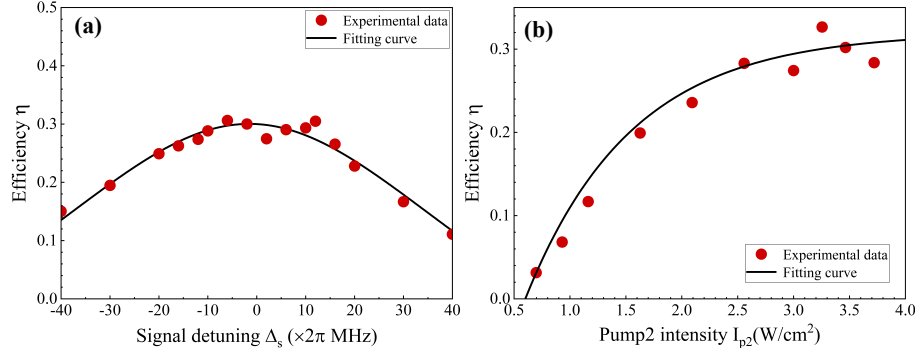


Fig. 2. (a) Measured frequency-conversion efficiency η as a function of signal detuning Δ_s . Red dot: experimental measurement. Black line: a Lorentzian fit. (b) The relation of frequency-conversion efficiency η to pump2 intensity. Red dot: experimental data. Black line: fitted curve.

We study the dependence of frequency-conversion efficiency on the pump2 intensity in Fig. 2(b), where the results are obtained under the optimal detuning conditions of $\Delta_s = -2\pi \times 6$ MHz and $\Delta_{p2} = -2\pi \times 36$ MHz, the mean signal photon per pulse is $\bar{n} = 0.8$. The power of pump1 P_1 is fixed at 1.87 mW and the pump2 intensity is calculated by the formula ($I_{p2} = P_2/(\pi w_{p2}^2)$), where $w_{p2} = 117 \mu\text{m}$ and P_2 are the beam waist and the power of pump2 respectively. Owing to the finite number of atoms and signal photons that interact with optical fields, we observe a tendency of saturation as the pump2 intensity increases. As shown in Fig. 2(b), the η grows with the increase of I_{p2} and a trend of saturation can be observed. This result is consistent with the conclusion drawn in the previous work [20]. To visualize the rising trend of the η , we fit the experimental data with $\eta = \eta_{\max}(1 - \exp(-(I_{p2} - I_0)/I_s))$ as proposed by [20], where $\eta_{\max} = 0.32$, $I_0 = 0.6 \text{ W/cm}^2$ and $I_s = 0.95 \text{ W/cm}^2$. The maximum η achieved in our system is 32%.

A crucial factor to achieve high frequency-conversion efficiency is the OD of our ⁸⁵Rb ensemble. We experimentally investigate the influences of different OD on conversion efficiency η at the conditions of $\Delta_s = -2\pi \times 6$ MHz, $\Delta_{p2} = -2\pi \times 36$ MHz, $P_1 = 1.87 \text{ mW}$, $P_2 = 1.4 \text{ mW}$ and $\bar{n} = 0.8$, which is illustrated in Fig. 3 (blue dots). The OD is defined as $OD = \rho\sigma L$ [21], where L is the length of atomic cloud, ρ is the number density of atoms, and $\sigma = 3\lambda^2/(4\pi)$ is the resonant absorption cross-section. Therefore an increase in OD indicates an increase in the number density of atoms, which implies an enhanced probability of FWM process occurring between optical fields and atoms. We can fetch the information from Fig. 3 (blue dots) that the increase in this probability has a significant influence on the improvement of the η . Considering the specific relationship between OD and η , we refer to the theoretical analysis in previous work [21]. The co-moving propagation equation for signal (E_s^+) and telecom (E_t^+) fields under momentum and energy conservation conditions are described as

$$\begin{aligned} \frac{d}{dz}E_s^+ &= \kappa_s E_t^+ + \alpha_s E_s^+, \\ \frac{d}{dz}E_t^+ &= \beta_t E_t^+ + \kappa_t E_s^+, \end{aligned} \quad (2)$$

where the self-coupling coefficients α_s , β_t and parametric coefficients κ_s , κ_t are defined by H.H.Jen and T.A.B.Kennedy in Ref. [21]. The self-coupling coefficient α_s (β_t) represents the self-gain of the signal (telecom) field transmitted in the medium, and the parametric coefficient κ_t (κ_s) represents the conversion gain from the signal (telecom) field to telecom (signal) field in the FWM process. The conversion efficiency η under conditions of frequency conversion with boundary condition $E_t^+(0) = 0$ can be defined by

$$\eta = \left| \frac{E_t^+(L)}{E_s^+(0)} \right|^2 = \left| \frac{\kappa_t}{2\omega} e^{(\alpha_s + \beta_t)OD/(2\rho\sigma)} (e^{\omega OD/(\rho\sigma)} - e^{-\omega OD/(\rho\sigma)}) \right|^2 \quad (3)$$

with $\omega = \sqrt{q^2 + \kappa_s \kappa_t}$, $q = (\beta_t - \alpha_s)/2$. We fit our experimental data with the Eq. (3), which is shown in Fig. 3 (black curve). The result is in good agreement with our experimental data and indicates that an atomic ensemble with a large OD is accompanied by a high frequency-conversion efficiency. One of the primary factors that limits the maximum frequency-conversion efficiency of our system is the OD of the atomic ensemble.

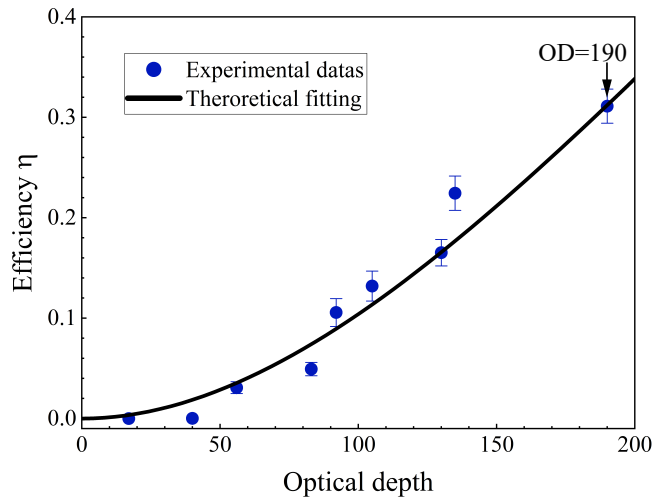


Fig. 3. Experimental data (blue dot) and theoretical fitting (black curve) of frequency-conversion efficiency η with the growth of OD, and all data have the background noise subtracted. The error bar is estimated from Poissonian statistics and using Monte Carlo simulations.

Figure 4 indicates that the frequency-conversion efficiency η with different $\bar{n}$ of input signal field. The $\bar{n}$ is scanned from 0.06 to 3.1. We find that the η is independent of the $\bar{n}$, and the average η is 32%. In addition, due to the noise caused by the dark count of the detector and the unfiltered pump fields, the SNR of the detected telecom fields decreases with the reduction of the $\bar{n}$, and the SNR is higher than 10 when $\bar{n}$ is 0.2 in our experimental system. Our scheme performs well when $\bar{n}$ is at single photon level.

Compared to the previous work [13] that adopts a ladder configuration, in where exists difficulty filtering out the weak FWM light from the strong homo-chromatic pump light, the proposed converter can operate at single-photon level with a high SNR since the pump and FWM lights are hetero-chromatic. What's more, in contrast to other works [13,20], our setup with a co-linear structure enhances the probability of FWM process by increasing the overlapped area between optical fields and atomic ensemble.

We can conclude from Fig. 3 that one of the most effective ways to improve conversion efficiency in further is increasing the OD of the ensemble. Fortunately, there are several feasible

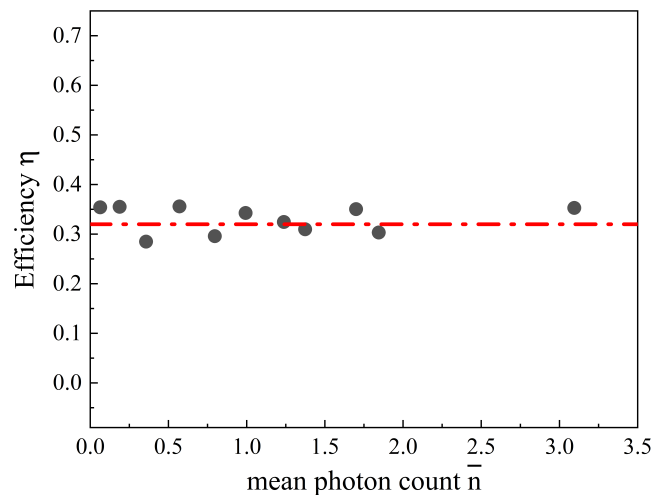


Fig. 4. Measured frequency-conversion efficiency η with the growth of $\bar{n}$ and all results have the back-ground noise subtracted.

methods towards this goal such as building a temporal dark spontaneous-force optical trap [24] or placing the atomic ensemble in a ring cavity [25] to increase the interaction length between optical fields and non-linear medium.

4. Conclusion

In conclusion, we report a telecom-wavelength (near C-band) frequency converter in cold ^{85}Rb ensemble with high OD through the FWM process. The frequency-down conversion at the wavelength from 795 nm to 1529.3 nm is achieved by a diamond-type energy level configuration. What's more, we experimentally confirm that the η improves significantly with the increase of OD, and the experimental data is in reasonable agreement with our theoretical fitting. Besides, we observe that the SNR in our work is well performed when $\bar{n}$ is at single photon level. Our scheme has the potential for applications in atom-based quantum repeater and long-distance quantum networks.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. A. Yariv and D. M. Pepper, "Amplified reflection, phase conjugation, and oscillation in degenerate four-wave mixing," *Opt. Lett.* **1**(1), 16–18 (1977).
2. D.-S. Ding, Z.-Y. Zhou, B.-S. Shi, X.-B. Zou, and G.-C. Guo, "Generation of non-classical correlated photon pairs via a ladder-type atomic configuration: theory and experiment," *Opt. Express* **20**(10), 11433–11444 (2012).
3. R. Willis, F. Becerra, L. Orozco, and S. Rolston, "Photon statistics and polarization correlations at telecommunications wavelengths from a warm atomic ensemble," *Opt. Express* **19**(15), 14632–14641 (2011).
4. R. Slusher, L. Hollberg, B. Yurke, J. Mertz, and J. Valley, "Observation of squeezed states generated by four-wave mixing in an optical cavity," *Phys. Rev. Lett.* **55**(22), 2409–2412 (1985).
5. J. Wu, Y. Liu, D.-S. Ding, Z.-Y. Zhou, B.-S. Shi, and G.-C. Guo, "Light storage based on four-wave mixing and electromagnetically induced transparency in cold atoms," *Phys. Rev. A* **87**(1), 013845 (2013).

6. A. J. Merriam, S. Sharpe, M. Shverdin, D. Manuszak, G. Yin, and S. Harris, "Efficient nonlinear frequency conversion in an all-resonant double- λ system," *Phys. Rev. Lett.* **84**(23), 5308–5311 (2000).
7. R. C. Pooser, A. M. Marino, V. Boyer, K. M. Jones, and P. D. Lett, "Low-noise amplification of a continuous-variable quantum state," *Phys. Rev. Lett.* **103**(1), 010501 (2009).
8. L.-M. Duan, M. D. Lukin, J. I. Cirac, and P. Zoller, "Long-distance quantum communication with atomic ensembles and linear optics," *Nature* **414**(6862), 413–418 (2001).
9. T. Chaneliere, D. Matsukevich, S. Jenkins, T. Kennedy, M. Chapman, and A. Kuzmich, "Quantum telecommunication based on atomic cascade transitions," *Phys. Rev. Lett.* **96**(9), 093604 (2006).
10. S. Shi, M.-X. Dong, Y.-C. Yu, Y.-H. Ye, W. Zhang, K. Wang, G.-C. Guo, D.-S. Ding, and B.-S. Shi, "Entangled qutrits generated in four-wave mixing without post-selection," *Opt. Express* **28**(8), 11538–11547 (2020).
11. D.-S. Ding, Z.-Y. Zhou, B.-S. Shi, X.-B. Zou, and G.-C. Guo, "Linear up-conversion of orbital angular momentum," *Opt. Lett.* **37**(15), 3270–3272 (2012).
12. D.-S. Ding, Z.-Y. Zhou, W. Huang, B.-S. Shi, X.-B. Zou, and G.-C. Guo, "Experimental up-conversion of images," *Phys. Rev. A* **86**(3), 033803 (2012).
13. D.-S. Ding, Z.-Y. Zhou, B.-S. Shi, and X.-B. Zou, "Efficient infrared upconversion via a ladder-type atomic configuration," *J. Mod. Opt.* **59**(20), 1768–1771 (2012).
14. G. Wang, Y. Xue, J.-H. Wu, Z.-H. Kang, Y. Jiang, S.-S. Liu, and J.-Y. Gao, "Efficient frequency conversion induced by quantum constructive interference," *Opt. Lett.* **35**(22), 3778–3780 (2010).
15. C.-Y. Lee, B.-H. Wu, G. Wang, Y.-F. Chen, Y.-C. Chen, and A. Y. Ite, "High conversion efficiency in resonant four-wave mixing processes," *Opt. Express* **24**(2), 1008–1016 (2016).
16. R. Willis, F. Becerra, L. Orozco, and S. Rolston, "Four-wave mixing in the diamond configuration in an atomic vapor," *Phys. Rev. A* **79**(3), 033814 (2009).
17. R. Kumar, V. Gokhroo, K. Deasy, and S. N. Chormaic, "Autler-townes splitting via frequency up-conversion at ultralow-power levels in cold rb 87 atoms using an optical nanofiber," *Phys. Rev. A* **91**(5), 053842 (2015).
18. C.-Y. Cheng, Z.-Y. Liu, P.-S. Hu, T.-N. Wang, C.-Y. Chien, J.-K. Lin, J.-Y. Juo, J.-S. Shiu, I. A. Yu, Y.-C. Chen, and Y.-F. Chen, "Efficient frequency conversion based on resonant four-wave mixing," *Opt. Lett.* **46**(3), 681–684 (2021).
19. A. Napoli, N. Calabretta, J. K. Fischer, N. Costa, S. Abrate, J. Pedro, V. Lopez, V. Curri, D. Zibar, E. Pincemin, S. Grot, G. Roelkens, C. Matrakidis, and W. Forysiak, "Perspectives of multi-band optical communication systems," in *The 23rd Opto-Electronics and Communications Conference*, (Institute of Electrical and Electronics Engineers, Technical Digest, 2018), pp. 1–2.
20. A. Radnaev, Y. Dudin, R. Zhao, H. Jen, S. Jenkins, A. Kuzmich, and T. Kennedy, "A quantum memory with telecom-wavelength conversion," *Nat. Phys.* **6**(11), 894–899 (2010).
21. H.-H. Jen and T. Kennedy, "Efficiency of light-frequency conversion in an atomic ensemble," *Phys. Rev. A* **82**(2), 023815 (2010).
22. S. Zhang, J. Chen, C. Liu, S. Zhou, M. Loy, G. K. L. Wong, and S. Du, "A dark-line two-dimensional magneto-optical trap of 85rb atoms with high optical depth," *Rev. Sci. Instrum.* **83**(7), 073102 (2012).
23. D.-S. Ding, W. Zhang, Z.-Y. Zhou, S. Shi, B.-S. Shi, and G.-C. Guo, "Raman quantum memory of photonic polarized entanglement," *Nat. Photonics* **9**(5), 332–338 (2015).
24. C. Townsend, N. Edwards, K. Zetie, C. Cooper, J. Rink, and C. Foot, "High-density trapping of cesium atoms in a dark magneto-optical trap," *Phys. Rev. A* **53**(3), 1702–1714 (1996).
25. C. Yang, S.-L. Liu, Z.-Y. Zhou, Y. Li, Y.-H. Li, S.-K. Liu, Z.-H. Xu, G.-C. Guo, and B.-S. Shi, "Extra-cavity-enhanced difference-frequency generation at 1.63 μm ," *J. Opt. Soc. Am. B* **37**(5), 1367–1371 (2020).